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The Influence of CEO Financial Background on Corporate Financialization

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Corresponding Author:	Jiayan Li, Ph.D. Guangdong University of Foreign Studies Guangzhou, Guangdong CHINA
First Author:	Dan Kang
Order of Authors:	Dan Kang
	Jiahao Xu
	Qiye Tong
	Jiayan Li, Ph.D.
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Title page

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Dan Kang

Department of Finance, Dongbei University of Finance and Economics, Dalian 116025, China

Email: kangdan2024@163.com

Jiahao Xu

School of Finance, Jiangxi Normal University, Nanchang 330022, China

Email: xujiahao112003@outlook.com

Qiye Tong

School of Business, Nankai University, Tianjin 300000, China

Email: tongqiye@outlook.com

Jiayan Li*

School of Accounting, Guangdong University of Foreign Studies, Guangzhou 510006, China

Email: 202210010@oamail.gdufs.edu.cn

*Corresponding author. Jiayan Li.

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Abstract

Entity enterprises are increasingly engaging in financial activities to enhance financial capital allocation. As core decision-makers, CEOs significantly impact the financialization of these enterprises. This paper, grounded in Top Echelon theory and Imprinting theory, examines the effect of a CEO's financial background on corporate financialization. Utilizing data from A-share listed companies from 2008 to 2022, the study aims to determine the direction and magnitude of this influence and explore the mediating roles and heterogeneity of these effects. Key findings include: (1) CEOs with a financial background significantly promote corporate financialization; (2) this promotion occurs primarily through increased managerial overconfidence and alleviated corporate financing constraints; and (3) the impact is more pronounced in non-state-owned and low-cash-flow firms compared to state-owned and high-cash-flow firms. Based on these insights, the paper suggests targeted management strategies.

Keywords: CEO financial background; corporate financialization; overconfidence; financing constraints

1. Introduction

The background characteristics of business leaders, particularly their financial background, profoundly impact the strategic decision-making and operational performance of enterprises (Oradi et al., 2020). As the core decision-maker of an enterprise, a CEO's financial background not only represents their professional ability and experience but also influences the enterprise's behavior and performance in financial markets (Liu et al., 2020). Understanding the significance of a CEO's financial background is crucial for comprehending corporate decision-making mechanisms, optimizing corporate governance structures, and enhancing market competitiveness. CEOs with a financial background possess profound knowledge of financial theory and rich practical experience, enabling them to make more informed and precise decisions when faced with complex financial problems. These CEOs can better grasp dynamic financial market changes, identify potential risks and opportunities, and thereby create more value for the enterprise (Yang et al., 2021). Furthermore, a financial background helps CEOs improve the financial management level and operational efficiency of the enterprise, focusing on capital liquidity, cost control, and capital allocation to manage financial resources more effectively and enhance market competitiveness.

The financialization of entity enterprises refers to the increasing participation of enterprises in financial market activities and the strategic use of financial capital (Liu et al., 2022). This trend reflects a growing understanding and utilization of financial capital by enterprises and the closer integration between finance and the real economy. By engaging in financial market activities, enterprises can leverage financial mechanisms to achieve capital appreciation and risk diversification. This financialization also promotes innovation and development within the financial market, fostering a positive interaction between financial capital and the real economy (Li et al., 2021). However, the financialization of entity enterprises also introduces risks and challenges, such as balancing financial investment with core business activities, controlling financial risks, and maintaining sound enterprise operations. Addressing these challenges requires

in-depth analysis and effective responses from business leaders (Song & Wu, 2022). This necessitates a thorough investigation into how a CEO's financial background impacts the financialization of entity enterprises, providing valuable insights and references for corporate decision-makers.

This study aims to explore the impact of a CEO's financial background on the financialization of entity enterprises. Specifically, it uses panel data from A-share listed firms from 2008 to 2022 to conduct an empirical analysis based on a double fixed-effects model to test the causal relationship between a CEO's financial background and the financialization of entity enterprises. The study also explores potential mediating roles of managerial overconfidence and financing constraints in this relationship, examining the mechanisms through which a CEO's financial background influences enterprise financialization. Additionally, it analyzes the heterogeneity of these effects from the perspectives of enterprise nature and cash flow, providing comprehensive insights into real-world complexities. Based on the empirical results, the paper proposes targeted development strategies.

The contribution of this paper is multi-faceted. First, it addresses a relative gap in current research on executive financial backgrounds, which often focuses narrowly on banking backgrounds without a holistic view of executive financial backgrounds (Li et al., 2017). This paper fills this gap, enriching the research on executive financial backgrounds and offering new perspectives and ideas. Second, it refines existing research by focusing specifically on CEOs, who play a decisive role in firms, rather than generalizing across all executives. This focus eliminates noise from varying managerial authorities and enhances the relevance and practicality of the findings. Third, the paper not only examines the impact of a CEO's financial background on corporate financialization but also reveals the underlying mechanisms and specific environments influencing this impact. This comprehensive analysis enriches the field of financialization and provides practical references for enterprises. Lastly, the paper expands the application scope of branding theory by exploring its relevance to the relationship between a CEO's financial background and corporate financialization, thus enhancing the theoretical depth and breadth of the study.

The structure of this paper is organized as follows: Section 2 provides a literature review, summarizing existing research related to the topic. Section 3 details the methodology and data used in the study. Section 4 presents the results and discussion, offering an analysis of the empirical findings. Finally, Section 5 concludes the paper, summarizing the main insights and proposing future research directions.

2. Literature review and research hypothesis

Top echelon theory emphasizes the significant impact of the traits and background of a firm's top management team (TMT) on the firm's strategic decisions and performance (Hambrick and Mason, 2004) Among them, the CEO, as a core member of the top management team, has a non-negligible role to play in the development of the enterprise in terms of his or her traits and background characteristics. Especially in the context of financialization of entity enterprises, the impact of CEO's financial background on the process of financialization of firms becomes more and more prominent (Hambrick, 2007). First of all, in the process of financialization of entity enterprises, a large number of financial activities are involved, such as financing, investment, risk

management, etc., which all require professional financial knowledge and skills. Therefore, CEOs with financial backgrounds can better grasp the dynamics of the financial market, identify potential risks and opportunities, and provide strong support for the financialization process of enterprises (Carpenter et al., 2004). Second, the CEO's financial background helps to enhance the financialization ability of the enterprise, which refers to the enterprise's ability to use the financial market for resource allocation and risk management. CEOs with financial backgrounds can establish closer cooperative relationships with financial institutions, obtain more financial resources and information support, and further promote the financialization process of enterprises (Wang et al., 2016). In addition, the financial background of the CEO also helps optimize the financialization strategy of the enterprise. Financialization strategy is the financialization development goal and action plan formulated by the enterprise according to the market environment and the enterprise's conditions (Waldman et al., 2004). CEOs with a financial background can more accurately grasp the changing trends of the financial market, combine them with the actual situation of the enterprise, and make comprehensive consideration of the enterprise's resource allocation, risk control, and other aspects from the perspective of finance to ensure that the enterprise maintains a sound development trend in the process of financialization (Whitler et al., 2021).

Imprinting theory has its roots in biology. Both individuals and organizations have several environmentally sensitive periods in their developmental process (Patten et al., 2014). The environmental characteristics of a particular phase will have a significant impact on the individual or organization. Individuals and organizations will be imprinted with environmental characteristics, and this imprint will continue to affect the behavior of individuals and organizations, and will not disappear easily (Marquis and Tilcsik, 2013). The financial background of a CEO leaves a deep imprint on his or her career, which may originate from his or her early work experience in a financial institution or through the systematic study of financial literacy and practical experience accumulation (Mathias et al., 2015) These experiences give CEOs an in-depth understanding and perception of financial markets, financial instruments, and risk management, forming a unique financial cognitive stigma. On the one hand, CEOs' financial background stigma influences their affective tendencies and risk preferences. CEOs with financial backgrounds usually hold a more open and positive attitude toward the financial market and are more willing to try new financial instruments and strategies (Shan et al., 2024). At the same time, they also tend to be more cautious and robust in risk control. This emotional stigma and risk preference stigma enable CEOs to weigh the pros and cons more rationally in the process of promoting the financialization of entity enterprises and formulate a financialization strategy that is in line with the actual situation of the firm (Shan et al., 2024). On the other hand, the stigma of CEOs' financial backgrounds is also reflected in their behavioral decisions. CEOs with financial backgrounds can use financial instruments more skillfully and conduct resource allocation and risk management more effectively when facing financialization decisions. They can optimize the financing structure of enterprises and improve the efficiency of capital use from a financial perspective, thus promoting the financialization process of entity enterprises (Cheng et al., 2023). Combined with the above analysis, this paper proposes research hypothesis 1.

H1: The CEO's financial backgrounds significantly contribute to the financialization of entity enterprises.

A CEO's financial background provides him or her with deep knowledge of financial theory and rich practical experience. This professional knowledge and experience make CEOs more confident in analyzing and judging the financial market, and thus more likely to develop an overconfident psychological state (Chen et al., 2024). Overconfident CEOs tend to overestimate their ability and control over the market and underestimate potential risks, thus showing higher risk-taking and risk appetite in financial investment decisions. They are more inclined to adopt aggressive financial investment strategies and pursue higher financial returns, thus accelerating the financialization process of entity enterprises (Zhong and Yan, 2024). Second, the financial background of CEOs helps them to establish extensive financial networks and resources.CEOs have access to more financial resources and information support, which further enhances their overconfidence and makes them more likely to adopt risky strategies in pursuit of higher financial returns (Zhong and Yan, 2024)[Error! Bookmark not defined.]. In addition, CEOs with a financial background are usually better able to understand and capitalize on the dynamic changes in financial markets, from which they can capture opportunities and avoid risks (Lun, 2022). However, this ability may also lead CEOs to be overconfident and ignore market feedback and warning signals. They may have too much faith in their own judgment and decision-making ability, and thus make overly risky decisions during the financialization process (Jiang et al., 2022). This overconfidence may expose firms to greater risks and challenges in the pursuit of financial gains. In summary, CEOs' financial background increases managers' overconfidence by providing deep financial knowledge, building extensive financial networks and resources, and coping with the complexity of financial markets, and this overconfidence promotes the financialization process of entity enterprises. Accordingly, research hypothesis 2 is proposed.

H2: The CEO's financial background significantly contributes to the financialization of entity enterprises by increasing managerial overconfidence.

The CEO's financial background brings professional financial knowledge and practical experience to the enterprise, thus effectively alleviating the enterprise financing constraints. In a modern market economy, financing constraints are one of the key factors restricting the development of entity enterprises (Sisli, 2012). CEOs with financial backgrounds gain insight into financing opportunities and risks through their deep theoretical knowledge of finance and rich practical experience. They can choose appropriate financing channels and methods according to the actual situation and financial needs of the enterprise, to ensure that the enterprise can obtain sufficient financial support at the critical moment (Wu and Hu, 2021). In addition, CEOs with financial backgrounds can also provide professional financing advice and guidance to enterprises, helping them optimize the financing structure and further reduce financing costs. Secondly, the financial background of CEOs helps enterprises establish close cooperative relationships with financial institutions, thus alleviating financing constraints (Lun, 2022). As an important channel for capital financing, financial institutions play a crucial role in the development of entity enterprises. CEOs with financial backgrounds understand the operation mechanism and risk preferences of financial institutions and can better communicate and coordinate with financial institutions. By establishing mutual trust with financial institutions, enterprises can obtain more financing support and preferential policies, thus effectively alleviating financing constraints (Gu et

al., 2023). In addition, the CEO's financial background helps enterprises grasp the opportunities of the financial market and achieve the optimal allocation of financial assets, thus alleviating financing constraints. The financial market is a market full of opportunities and challenges, and CEOs with financial backgrounds can discover and utilize investment opportunities in the financial market through their professional financial knowledge and keen insight. They can realize the financialized income of the enterprise by investing in financial assets such as stocks, bonds, funds, etc., and bringing additional sources of capital for the enterprise (Zhong et al., 2023). CEOs with financial backgrounds are also able to formulate a reasonable asset allocation plan according to the risk tolerance and return requirements of the enterprise to achieve the optimal allocation of financial assets and further reduce the financing costs and risks of the enterprise (Ding et al., 2023). In summary, the financial background of CEOs effectively alleviates the financing constraints of enterprises and promotes the level of financialization of entity enterprises by providing professional financial knowledge and practical experience, establishing close cooperation with financial institutions, and grasping financial market opportunities. Accordingly, research hypothesis 3 is proposed.

H3: The financial background of CEOs significantly contributes to the financialization of entity enterprises by easing financing constraints.

3. Research design

3.1 Model specifications

Referring to the study of Chen et al. (2024), this paper constructs the following double fixed-effects model as the benchmark model of this paper, and all subsequent empirical tests will be carried out based on this model.

$$FINRATIO_{it} = \alpha_0 + \alpha_1 CEOFIN_{it} + \alpha_2 \boldsymbol{Controls}_{it} + \theta_i + \sigma_t + \varepsilon_{it} \quad (1)$$

In model (1), i denotes the ith bank, and t denotes the tth year.$FINRATIO_{it}$ are the core explanatory variables of the paper, which are proxies for the financialization of entity enterprises, meaning the loan size and risk of the ith commercial bank in year t. α_0 is the constant term of the model. $CEOFIN_{it}$ is the core explanatory variable of the paper, a proxy variable for the Financial background of CEOs. α_1 is the $CEOFIN_{it}$ the corresponding coefficients to be estimated. $\boldsymbol{Controls}_{it}$ are a set of control variables selected for the paper. In addition, the paper incorporates firm-level individual fixed effects$\theta_i$and year-level time fixed effectsσ_t. ε_{it} are unobservable random disturbance terms. Where α_1 is the core estimated coefficient of interest in this paper, the expected α_1 direction is positive, i.e., the CEO's financial background promotes the level of financialization of entity enterprises.

Baron et al. (1986) provide a more complete description and introduction of the mediation effect model, proposing the use of a three-step approach to realize the test of the mediation mechanism. This paper draws on the ideas of Baron et al. (1986) to propose the following three-step mediation effect model: first, test the impact of CEO's financial background on the financialization of entity enterprises, which has been tested in model (1); second, test the impact of CEO's financial background on the intermediary variable, which is tested through model (2); third, put the CEO's financial background and the intermediary variable into the model at the same time as the independent variable and financialization of entity enterprises as dependent variables, focusing on the effect of mediating variables on financialization of entity enterprises, which is

tested through model (3). If γ_1 andδ_2 are significant at the same time, it indicates that the mediating effect exists.

$$M_{it} = \gamma_0 + \gamma_1 CEOFIN_{it} + \gamma_2 \boldsymbol{Controls}_{it} + \theta_i + \sigma_t + \varepsilon_{it} \quad (2)$$

$$FINRATIO_{it} = \delta_0 + \delta_1 CEOFIN_{it} + \delta_2 M_{it} + \delta_3 \boldsymbol{Controls}_{it} + \theta_i + \sigma_t + \varepsilon_{it} \quad (3)$$

3.2 Variable definition

The dependent variable in this paper is the level of financialization of entity enterprises, which reflects the allocation and operation of financial assets in the business process of entity enterprises (Zheng et al., 2019). A higher level of financialization may mean that enterprises focus more on obtaining revenue through financial activities rather than through traditional production and operation activities. Too high a level of financialization may mean that enterprises are exposed to greater financial risks, which may have a greater impact on the stable operation of enterprises once the financial market fluctuates (Xu and Guo, 2023). Especially when entity enterprises allocate a large amount of funds to the stock market, real estate, and other financial areas to chase short-term excess returns, the level of financialization of entity enterprises will rise, which may lead to the hollowing out of the real economy and hurt the country's economic development (Song and Wu, 2022). In this paper, referring to the study of Qi et al. (2021), the proportion of financial assets to total assets is used as a proxy variable for the financialization of entity enterprises, and the specific formula is as follows:

FINRATIO = (Trading financial assets + Derivative financial assets + Net loans and advances granted + Net available-for-sale financial assets + Net held-to-maturity investments + Net investment properties) / Total assets (4)

The independent variable in this paper is CEO's financial background, which mainly refers to the CEO's experience and engagement in financial institutions or related fields (Lun, 2022). Specifically, a CEO can be said to have a financial background if he or she has held positions in policy banks, commercial banks, investment banks, financial regulators, fund management companies, insurance companies, exchanges, securities firms, securities registration and settlement firms, futures firms, trust companies, investment management firms, and other financial institutions (Oradi et al., 2020). This paper uses a dummy variable for whether the CEO has a financial background as a proxy variable for the CEO's financial background, which is 1 if the CEO has a financial background and 0 if the CEO does not have a financial background (Jiang et al., 2022).

The mediating variables in this paper are overconfidence and financing constraints. Management overconfidence refers to managers' overconfidence in their abilities and judgments in the decision-making process, holding an overly optimistic attitude toward the future development of the enterprise, market competition, and risk assessment (Russo and Schoemaker, 1992). This overconfidence may lead enterprises to make irrational, even high-risk choices in conducting strategic planning, investment decisions, and market expansion (Adam et al., 2015). The study (Kramer and Liao, 2016), which this paper refers to, constructs the overconfidence variable through the following procedure: first, taking the growth rate of total assets as the dependent variable and the growth rate of operating income as the independent variable, regression analysis is carried out to obtain the corresponding residuals of each enterpriseε, as shown in model (4). Secondly, if the residuals of enterprises ε are larger than the industry residual ε of the median, then the enterprise has managerial overconfidence, and managerial confidence takes the value of 1,

otherwise it is 0.

$$GORWTH_ROA_{it} = \theta_0 + \theta_1 GROWTH_REVE_{it} + \theta_i + \sigma_t + \varepsilon_{it} \quad (5)$$

Financing constraints refer to the various limitations and constraints that enterprises are subjected to in the process of financing, which may come from financing channels, financing methods, financing costs, and the amount of financing (Hadlock et al., 2010). These constraints make it difficult for enterprises to raise the required funds according to their wishes, which may increase their financing costs and even lead to a financial crisis. This paper uses the SA index as a proxy variable for corporate financing constraints (Xu et al., 2023).

Referring to the studies of Gao et al. (2023) and Qi et al. (2021), this paper selects firm size, gearing ratio, return on net assets, current ratio, intangible assets ratio, capital accumulation ratio, financial leverage, board of directors' size, percentage of independent directors, and two-vocations-at-one-place as the control variables to exclude the interference of other factors as much as possible and alleviate the endogeneity problem. The variables are selected and defined as shown in the table below.

Table 4-2 Variable Selection and Definition

typology	variable name	Variable Abbreviations	Variable definition
explanatory variable	Financialization of entity enterprises	*FINRATIO*	See model (4)
explanatory variable	CEO's financial background	*CEOFIN*	1 if the CEO has a financial background and 0 if the CEO does not have a financial background
intermediary variable	overconfidence	*OC*	See model (5)
	Financing constraints	*SA*	SA index
control variable	Enterprise size	*Size*	Logarithmic value of total assets
	gearing	*Lev*	Total liabilities/total assets
	return on net assets	*ROE*	Profit after tax/net assets
	current ratio	*Liquid*	Total current assets/total current liabilities
	Intangible assets as a percentage	*Intangible*	Total intangible assets/total assets
	rate of capital accumulation	*RCA*	Increase in owners' equity for the year/owners' equity at the beginning of the year
	financial leverage	*FL*	Change in earnings per ordinary share/change in EBITDA
	Board size	*Board*	Logarithmic value of the number of board members
	Percentage of independent directors	*Indep*	Number of independent directors/total number of board members
	two jobs in one	*Dual*	Chairman and general manager of both positions is 1, otherwise 0

3.3 Data sources

The firm-level data used in this paper comes from the Cathay Pacific database. After excluding financial firms, ST or PT, no delisted and insolvent firms during the sample period, 24,004 valid samples are obtained, encompassing 3,919 listed firms from 2008-2022.

In this paper, the acquired raw data are processed as follows. (1) Absolute value data with large magnitude are logarithmically processed to alleviate the problem of heteroskedasticity that may be brought about by large differences in magnitude. Such as enterprise size, etc. (2) A few samples with missing data in some years were filled in using linear interpolation. (3) Bilateral tail reduction at the 1% level is performed for all continuous variables to mitigate the interference caused by extreme values as much as possible. The final descriptive statistics of the full-sample variables obtained are shown in the table below.

Table 2 Table of descriptive statistics

VarName	Obs	Mean	SD	Min	Median	Max.
FINRATIO	24004	0.06	0.09	0.000	0.016	0.483
CEOFIN	24004	0.05	0.22	0.000	0.000	1.000
OC	24004	0.50	0.50	0.000	0.000	1.000
SA	24004	-3.77	0.26	-4.370	-3.764	-3.153
Size	24004	22.20	1.30	19.872	21.996	26.000
Lev	24004	0.42	0.20	0.053	0.419	0.867
ROE	24004	0.07	0.11	-0.414	0.077	0.348
Liquid	24004	2.52	2.61	0.355	1.670	16.744
Intangible	24004	0.05	0.05	0.000	0.033	0.285
RCA	24004	0.16	0.36	-0.353	0.068	2.206
FL	24004	1.29	0.92	-0.382	1.049	6.528
Board	24004	2.14	0.20	1.609	2.197	2.639
Indep	24004	0.37	0.05	0.333	0.333	0.571
Dual	24004	0.26	0.44	0.000	0.000	1.000

4. Empirical results and analysis

4.1 Benchmark regression

To empirically test the impact of CEO's financial background on the financialization of entity enterprises, this paper uses STATA17 software to conduct subsequent regression and correlation tests. Covariance refers to the high degree of correlation that may exist between variables due to common trends or similar characteristics, etc., which can seriously interfere with the regression results (Basagaña et al., 2022) . To avoid multiple covariance interfering with the regression results, this paper calculated the variance inflation factor of each variable, and the results are shown in Table 3. It can be seen that the variable VIF in all models is less than the empirical value of 10, indicating that the model in this paper does not have the problem of covariance.

Table 3 Multiple covariance test

Variable	VIF	1/VIF
Lev	2.47	0.40
Liquid	1.91	0.52
Board	1.67	0.60
Size	1.55	0.64
Indep	1.53	0.65
ROE	1.43	0.70
RCA	1.24	0.80

FL	1.07	0.93
Dual	1.07	0.94
Intangible	1.04	0.96
CEOFIN	1.00	1.00
Mean VIF	1.45	

The regression results obtained after accounting for the individual bank as well as year-time effects are shown in Table 4 below. The coefficient of the effect of CEO's financial background on the financialization of entity enterprises without adding control variables in column (1) is 0.0060, which passes the test of significance at the 1% level, which indicates that the CEO's financial background significantly increases the level of financialization of entity enterprises. The goodness of fit of the model without adding control variables is 0.152, which is a good fit, indicating that the variables in the model explain the changes in the financialization of entity enterprises better. After adding control variables in column (1), the coefficient of the effect of CEO's financial background on the financialization of entity enterprises is 0.0074, which passes the test of significance at the 1% level, and the CEO's financial background still exhibits a significant positive effect on the level of financialization of entity enterprises. Hypothesis H1 is confirmed.

Table 4 Benchmark regression results

	(1)	(2)
	FINRATIO	FINRATIO
CEOFIN	0.0060***	0.0074***
	(2.7237)	(3.3608)
Size		-0.0097***
		(-9.4252)
Lev		-0.0586***
		(-12.2632)
ROE		-0.0185***
		(-4.0076)
Liquid		-0.0015***
		(-5.5128)
Intangible		-0.1562***
		(-10.5712)
RCA		0.0024**
		(2.0381)
FL		0.0012***
		(2.6505)
Board		-0.0023
		(-0.5302)
Indep		0.0130
		(1.0215)
Dual		-0.0009
		(-0.6652)
Firm FE	YES	YES
Year FE	YES	YES

_cons	0.0290***	0.2682***
	(17.1609)	(11.4484)
N	24004	24004
R^2	0.152	0.172

Note: *** p<0.01, ** p<0.05, * p<0.1, t-statistics in parentheses.

4.2 Robustness Tests

To ensure the validity of the regression results, this paper conducts robustness tests by replacing the explanatory variables, adjusting the research window period, lagging the dependent variable by one period, and PSM-DID, respectively. Specifically, this paper uses the dummy variable of whether entity enterprises have made financial investments (FINRATIO2) as a proxy variable for the financialization of entity enterprises, and the results are shown in column (1) of Table 5. The study window period is adjusted to 2008-2019 to exclude the interference of the new crown epidemic, and the results are shown in column (2) of Table 5. The dependent variable is lagged by one period to capture the lagged impact of the CEO's financial background on the financialization of entity enterprises, and the results are shown in column (3) of Table 5. 4 Tight-neighbor matching is performed with the CEO's financial background as the treatment variable, and then PSM-DID regression analysis is carried out, and the results are shown in column (4) of Table 5. Overall, the robustness test results maintain good agreement with the benchmark regression.

Table 5 Robustness test

	(1) Substitution of explanatory variables	(2) Adjusting the research window	(3) Dependent variable lagged one period	(4) PSM-DID
	FINRATIO2	FINRATIO	FINRATIO	FINRATIO
CEOFIN	0.0186*	0.0070***		0.0057
	(1.6603)	(2.9574)		(1.5048)
LCEOFIN			0.0088***	
			(3.4274)	
Size	0.0411***	-0.0073***	-0.0107***	-0.0114***
	(7.8341)	(-6.4039)	(-8.6108)	(-4.0159)
Lev	0.0407*	-0.0442***	-0.0561***	-0.0297**
	(1.6684)	(-8.8359)	(-9.9127)	(-2.2286)
ROE	-0.0100	-0.0121**	-0.0238***	-0.0088
	(-0.4251)	(-2.4918)	(-4.4150)	(-0.6884)
Liquid	-0.0201***	-0.0019***	-0.0017***	0.0008
	(-14.0007)	(-6.9107)	(-4.8449)	(1.0429)
Intangible	-0.1055	-0.1462***	-0.1704***	-0.2145***
	(-1.3978)	(-9.5869)	(-9.7347)	(-5.0994)
RCA	-0.0021	0.0014	0.0049***	0.0022
	(-0.3503)	(1.2251)	(3.3013)	(0.6444)
FL	0.0042*	0.0001	0.0010*	0.0011
	(1.7874)	(0.2684)	(1.8783)	(0.7326)

Board	0.0197	0.0015	-0.0074	0.0087
	(0.9028)	(0.3350)	(-1.5099)	(0.7474)
Indep	0.0573	0.0197	0.0096	0.0500
	(0.8779)	(1.5017)	(0.6723)	(1.4490)
Dual	-0.0172**	-0.0000	-0.0002	-0.0074**
	(-2.4842)	(-0.0128)	(-0.1030)	(-2.1092)
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
_cons	-0.3044**	0.1999***	0.3112***	0.2595***
	(-2.5433)	(7.7228)	(11.0975)	(3.9651)
N	24004	17139	18179	5466
R^2	0.277	0.153	0.172	0.148

Note: *** p<0.01, ** p<0.05, * p<0.1, t-statistics in parentheses.

4.3 Intermediation mechanism test

This paper separately tests the possible mediating role of overconfidence and financing constraints in the impact of CEO's financial background on the financialization of entity enterprises, and the regression results are shown in Table 6. The coefficient of the effect of CEO's financial background on overconfidence in column (1) is 0.0455, which passes the significance test at the 1% level, indicating that CEO's financial background significantly increases the level of managerial overconfidence; the coefficient of the effect of overconfidence on the financialization of entity enterprises in column (2) is 0.0023, indicating that overconfidence significantly increases the level of financialization of entity enterprises. The benchmark regression together with the results in (1) and (2) indicate that the CEO's financial background increases the level of financialization of entity enterprises by significantly increasing overconfidence and hypothesis H2 is confirmed. Similarly, the coefficient of the effect of CEO's financial background on financing constraints in column (3) is -0.0076, which passes the significance test at the 1% level, indicating that CEO's financial background significantly reduces the financing constraints faced by firms; and the coefficient of the effect of financing constraints on the financialization of entity enterprises in column (4) is -0.0724, which suggests that financing constraints significantly reduce the level of financialization of entity enterprises. The benchmark regression together with the results in (3) and (4) indicate that the CEO's financial background increases the level of financialization of entity enterprises by significantly reducing the financing constraints of firms, and hypothesis H3 is confirmed.

Table 6 Mechanism tests

	(1)	(2)	(3)	(4)
	OC	FINRATIO	SA	FINRATIO
CEOFIN	0.0455***	0.0073***	-0.0076***	0.0068***
	(2.5798)	(3.3133)	(-3.1272)	(3.1186)
OC		0.0023***		
		(2.6036)		
SA				-0.0724***
				(-11.4295)
Size	0.0529***	-0.0098***	0.0049***	-0.0093***

	(6.4180)	(-9.5347)	(4.2652)	(-9.1073)
Lev	0.6801***	-0.0601***	-0.0097*	-0.0593***
	(17.7297)	(-12.4934)	(-1.8268)	(-12.4491)
ROE	0.8655***	-0.0205***	-0.0065	-0.0190***
	(23.3432)	(-4.3782)	(-1.2745)	(-4.1232)
Liquid	-0.0130***	-0.0015***	0.0051***	-0.0012***
	(-5.7425)	(-5.4036)	(16.2518)	(-4.1917)
Intangible	-0.4300***	-0.1552***	0.0890***	-0.1497***
	(-3.6248)	(-10.5027)	(5.4190)	(-10.1606)
RCA	0.3955***	0.0015	-0.0107***	0.0016
	(41.6788)	(1.2206)	(-8.1571)	(1.3842)
FL	-0.0323***	0.0013***	0.0007	0.0013***
	(-8.6741)	(2.8051)	(1.3383)	(2.7669)
Board	0.0792**	-0.0024	0.0010	-0.0022
	(2.3108)	(-0.5726)	(0.2174)	(-0.5143)
Indep	0.0619	0.0129	0.0277*	0.0150
	(0.6040)	(1.0105)	(1.9519)	(1.1822)
Dual	0.0122	-0.0009	0.0076***	-0.0004
	(1.1160)	(-0.6857)	(5.0146)	(-0.2625)
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
_cons	-1.0988***	0.2707***	-3.6009***	0.0077
	(-5.8398)	(11.5476)	(-138.2695)	(0.2353)
N	24004	24004	24004	24004
R^2	0.173	0.172	0.853	0.177

Note: *** p<0.01, ** p<0.05, * p<0.1, t-statistics in parentheses.

4.4 Heterogeneity analysis

Entity firms are classified into state-owned and non-state-owned firms to examine how different firm natures would make the effect of CEO's financial background on the financialization of entity firms exhibit heterogeneity, and the regression results are shown in columns (1) and (2) of Table 7. In the sample of state-owned enterprises, the coefficient of the impact of CEO's financial background on the financialization of entity enterprises is 0.0049, but it does not pass the significance test at any level; in the sample of non-state-owned enterprises, the coefficient of the impact of CEO's financial background on the financialization of entity enterprises is 0.0100, which passes the significance test at the 1% level. It can be seen that the impact of CEO's financial background on the financialization of entity enterprises is greater in non-state-owned firms compared to the sample of state-owned firms.

State-owned enterprises (SOEs) tend to be subject to more government intervention and policy guidance due to their special property rights structure and business objectives (Lin et al., 2020). Such intervention and guidance limit the CEO's autonomy in the decision-making process to a certain extent, especially in financial investment and capital operation (Helmich and Gilroy, 2012). Therefore, even if the CEO has a rich financial background, his or her influence on the financialization of the firm may be constrained to a certain extent. However, in non-state-owned

enterprises, CEOs have relatively greater decision-making power, and their business objectives are more inclined to maximize economic benefits (Zhou et al., 2017). When CEOs have a financial background, they are better able to utilize their professional knowledge and experience to grasp market opportunities and make financial investments and capital operations. In this context, the financial background of CEOs has a more obvious role in promoting the financialization of entity enterprises. In addition, the financing environment faced by non-state-owned enterprises is usually more severe, especially in obtaining bank loans and capital market financing (Tan et al., 2014). CEOs with financial backgrounds are more likely to optimize the resource allocation of enterprises and improve operational efficiency through financialization.

The cash flow status of the enterprise will be more limited to the play of the CEO's financial background and strength, thus making the impact of the CEO's financial background on the financialization of the entity enterprise show obvious heterogeneity. Based on the mean value of the enterprise's cash flow, entity enterprises are divided into high cash flow enterprises and low cash flow enterprises, and regression analysis is carried out separately, and the regression results are shown in columns (3) and (4) of Table 7. In the sample of high cash flow firms, the coefficient of the impact of CEO's financial background on the financialization of entity firms is 0.0049, but it does not pass the test of significance at any level; in the sample of low cash flow firms, the coefficient of the impact of CEO's financial background on the financialization of entity firms is 0.0054, which passes the test of significance at the 1% level. It can be seen that the effect of CEO's financial background on the financialization of entity enterprises is greater in the low cash flow firms compared to the high cash flow firms sample.

From the perspective of firms' financial health, high cash-flow firms usually have more abundant working capital and stable capital flows, and therefore have greater flexibility and autonomy in their financial investment decisions (Almeida et al., 2004). Low cash flow firms, on the other hand, tend to face capital shortages and financing pressures, and their financial investment decisions are more likely to be constrained by their capital position (Das et al., 2013). Against this background, the role of CEOs with a financial background in low-cash-flow firms is particularly important. With their profound financial knowledge and experience, they can more accurately assess the firm's capital situation and market environment, and formulate a financialization strategy that is more in line with the reality of the firm. They are not only able to utilize the limited funds for investment more efficiently, but also able to improve the capital status of the enterprise and alleviate the financing pressure through reasonable capital operation (Sheikh, 2022).

Table 7 Heterogeneity analysis

	(1)	(2)	(3)	(4)
	nationalized business	non-state enterprise	high cash flow	low cash flow
	FINRATIO	FINRATIO	FINRATIO	FINRATIO
CEOFIN	0.0049	0.0100***	0.0049	0.0054**
	(1.6326)	(3.2910)	(0.9392)	(2.4182)
Size	-0.0051***	-0.0129***	-0.0089***	-0.0071***
	(-3.5331)	(-8.7775)	(-3.0401)	(-6.8521)
Lev	-0.0767***	-0.0456***	-0.0654***	-0.0559***
	(-11.4166)	(-6.8179)	(-4.7322)	(-11.2870)
ROE	-0.0325***	-0.0070	-0.0012	-0.0272***

	(-5.2043)	(-1.0668)	(-0.0946)	(-6.0338)
Liquid	-0.0010	-0.0010***	-0.0003	-0.0033***
	(-1.6266)	(-2.8788)	(-0.4972)	(-6.4801)
Intangible	-0.1183***	-0.1683***	-0.1908***	-0.1550***
	(-6.1782)	(-7.7882)	(-4.2951)	(-10.6298)
RCA	0.0054***	0.0013	-0.0008	0.0051***
	(3.2026)	(0.8171)	(-0.3264)	(3.8537)
FL	0.0010*	0.0015*	0.0059***	0.0008**
	(1.9283)	(1.9333)	(2.7784)	(2.0516)
Board	0.0001	-0.0073	-0.0227**	0.0058
	(0.0185)	(-1.1636)	(-2.0705)	(1.3694)
Indep	0.0105	0.0232	-0.0594*	0.0219*
	(0.6713)	(1.1888)	(-1.8092)	(1.7364)
Dual	0.0003	-0.0016	0.0015	-0.0017
	(0.1344)	(-0.9299)	(0.4423)	(-1.2071)
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
_cons	0.1840***	0.3208***	0.3065***	0.1946***
	(5.5625)	(9.6537)	(4.7234)	(8.2064)
N	9406	14598	7821	16183
R^2	0.146	0.193	0.298	0.107

Note: *** p<0.01, ** p<0.05, * p<0.1, t-statistics in parentheses.

5. Conclusions and implications

5.1 Conclusions

This paper focuses on the impact of CEO's financial background on entity enterprises financialization. The panel data of 3919 A-share listed enterprises from 2008-2022 is used to carry out empirical analyses, explore the causal relationship between the two, explicate mechanism, and examine the asymmetric influence under different conditions. The main research conclusions are as follows. (1) The CEO's financial background significantly increases the level of financialization of entity enterprises. This effect still holds significantly after multiple robustness tests of replacing the explanatory variables, adjusting the research window period, lagging the dependent variable by one period, and PSM-DID. (2) The CEO's financial background significantly increases the level of financialization of entity enterprises mainly through two paths: increasing managerial overconfidence and alleviating corporate finance constraints. (3) The effect of CEO's financial background on the financialization of entity enterprises is more pronounced in non-state-owned firms and low-cash-flow firms.

5.2 Policy implications

The findings of this paper point out that CEO's financial background significantly increases the level of financialization of entity enterprises. While making full use of the CEO's financial background to promote the financialization of entity enterprises, enterprises need to be alert to the risk of excessive financialization. Therefore, it is suggested that when selecting CEOs with financial backgrounds, enterprises should emphasize their understanding of and commitment to

the development of the main business of the entity, in addition to their financial expertise. In the decision-making process, CEOs should adequately weigh the relationship between financial investment and the development of the main business, and ensure that the financial activities serve the sound development of the main business, instead of over-pursuing the financial gains and neglecting the main business.

Enterprises should establish a scientific decision-making mechanism to avoid CEOs making overly risky financial investment decisions due to overconfidence. The CEO's decisions can be supervised and checked through the establishment of a decision-making advisory committee and the introduction of external expert consultants. At the same time, the construction of corporate culture should be strengthened to advocate rational and sound business concepts, to reduce the phenomenon of overconfidence from the source. In response to the mechanism of easing the role of corporate financing constraints, enterprises should make full use of the advantages of the CEO's financial background, strengthen cooperation with financial institutions, broaden financing channels, and reduce financing costs. At the same time, optimize the internal financing structure of the enterprise, improve the efficiency of capital use, and reduce the financing risk. In addition, enterprises can also improve their credit rating among financial institutions by establishing a credit rating system and improving the quality of information disclosure, to obtain more financing support.

For non-state-owned enterprises, they should adhere to their main business orientation, ensure that financial activities serve the sound development of their main business, and avoid the blind pursuit of financial gains. Enhancing the financial literacy of management will help to view the financialization process more rationally and avoid excessive speculation. At the same time, optimize the corporate governance structure, play the role of supervision and checks and balances, and ensure the science and compliance of CEO decision-making. For low-cash-flow enterprises, they should improve their cash-flow situation by optimizing financial management and improving operational efficiency. At the same time, it actively seeks cooperation with financial institutions and utilizes the advantage of the CEO's financial background to broaden financing channels and alleviate financing constraints. In addition, the enterprise can also consider attracting more capital into it by introducing strategic investors and implementing equity incentives to provide financial support for the financialization process of the enterprise.

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